Workshop construction plan

Synthetic workshop 1500

Longitudinal split wood

Overall assembled length: 1500.0 mm

Internal mouthpiece diameter: 30.0 mm

Internal bell diameter: 83.0 mm

Target note / frequency: C#2 / 69.3 Hz

Predicted passive resonances: 77.5 / 189.0 / 286.0 / 396.5 / 512.3 Hz

Minimum remaining wall: 8.0 mm; Radial overcut reserve: 1.0 mm

Prepared stock for each half: 2

Length x width x thickness: 1500.0 x 101.0 x 50.5 mm

Prepare two flat joining faces. Mark stations from the mouthpiece on both halves. Hollow a semicircular groove to the listed depth from the joining plane; mirror, dry-fit and align marks before gluing.

Stock dimensions are after splitting and flattening. Add your measured saw kerf, facing and end-trimming allowances to the original blank; these are not specified by this plan.

Marking stations - all dimensions in mm

From mouthpiece	Inside diameter	Hollowing depth per half	Width (min.)	Thickness (min.)
0.0 *	30.0	15.0	48.0	24.0
100.0	31.3	15.6	49.3	24.7
200.0	32.5	16.3	50.6	25.3
300.0	33.8	16.9	51.9	26.0
400.0	35.1	17.5	53.1	26.6
500.0	36.4	18.2	54.4	27.2
600.0	37.6	18.8	55.7	27.9
700.0	38.9	19.5	57.0	28.5
800.0	40.2	20.1	58.2	29.1
900.0	41.5	20.7	59.5	29.8
1000.0	42.7	21.4	60.8	30.4
1020.0 *	43.0	21.5	61.0	30.5
1100.0	49.7	24.8	67.7	33.9
1200.0	58.0	29.0	76.0	38.0
1300.0	66.3	33.2	84.4	42.2
1400.0	74.7	37.3	92.7	46.4
1500.0 *	83.0	41.5	101.0	50.5

* Mandatory geometry transition. Before - after gives both sides of a step.

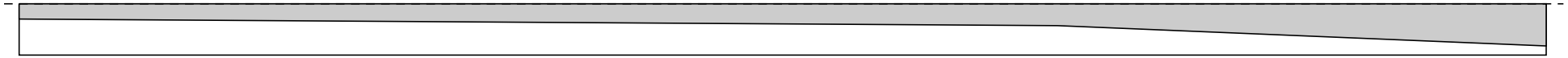
Passive acoustic prediction, not a guaranteed played note. Check material, joints and wall integrity.

Dimensioned construction drawing

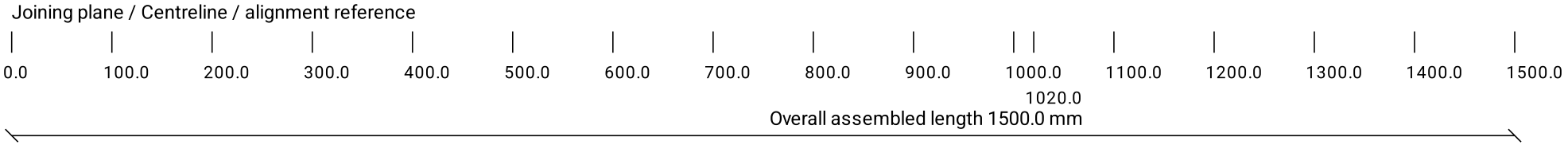
Synthetic workshop 1500 - Longitudinal split wood

Longitudinal scale 1:6.0

Half 1 - material to remove is shaded

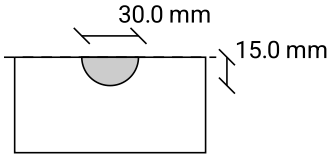


Half 2 - mirrored matching cavity



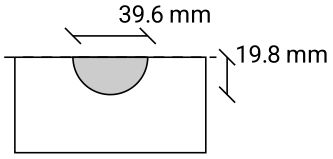
Semicircular grooves - section views - Shaded area: remove

From mouthpiece 0.0 mm



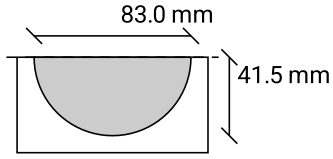
Cross-section scale 1:4.0

From mouthpiece 750.0 mm



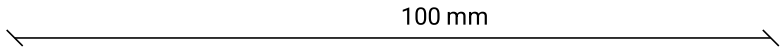
Cross-section scale 1:4.0

From mouthpiece 1500.0 mm



Cross-section scale 1:4.0

Dimensions rounded to 0.1 mm for marking; not a guaranteed manufacturing tolerance.



100 mm - print at 100% and verify with a ruler